Gravimetric Calculations
(CHE 2103)

Gravimetric Calculations

How to Perform a Successful Gravimetric Analysis

- **What steps are needed?**

1. Sampled dried, triplicate portions weighed
2. Preparation of the solution
3. Precipitation
4. Digestion
5. Filtration
6. Washing
7. Drying or igniting
8. Weighing
9. Calculation

Gravimetric Analysis

- Gravimetric Analysis – one of the most accurate and precise methods of macro-quantitative analysis.
- Analyte selectively converted to an insoluble form.
- Measurement of mass of material.
- Correlate with chemical composition

Gravimetric Analysis…..

- Precipitation Techniques
  - Add precipitating reagent to sample solution
  - Reacts with analyte to form insoluble material
  - Precipitate has known composition or can be converted to known composition
- Precipitate handling involves
  - Quantitative collection (no losses)
  - Isolation of pure product
- Measure mass of precipitate
- Calculation of original analyte content (concentration)

Gravimetric Analysis…..

- Desirable properties of analytical precipitates:
  - Readily filtered and purified
  - Low solubility, preventing losses during filtration and washing
  - Stable final form (unreactive)
  - Known composition after drying or ignition

Gravimetric Analysis…..

- Precipitating reagents:
- **Selective**
- Ag^+ + Halides (X^-) → $AgX_{(s)}$
- $Ag^+ + CNS^- \rightarrow AgCNS_{(s)}$
- **Specific**
- Dimethylglyoxime (DMG)
- $2\ DMG + Ni^{2+} \rightarrow Ni(DMG)_{2(s)} + 2\ H^+$

Gravimetric Analysis

- Calculations of analyte content requires knowledge of :
- Chemistry
- Stoichiometry
- Composition of precipitate

Gravimetry and Solution Equiliria

- Thermal Conversion to Measurable Form
- Removal of volatile reagents & solvent
- Extended heating at 110 to 115 ^{O}C
- Chemical conversion to known stable form
- $CaC_2O_{4(s)} \rightarrow CaO_{(s)} + CO_{(g)} + CO_{2(g)}$
- Volatilization & trapping of component
- $NaHCO_{3(aq)} + H_2SO_{4(aq)} \rightarrow CO_{2(g)} + H_2O + NaHSO_{4(aq)}$

Gravimetric Calculations

- **The point here is to find the weight of analyte from the weight of precipitate.**
- **Assume Cl_2 is to be precipitated as AgCl, then we can write a stoichiometric factor reading as follows:**
- **One mole of Cl_2 gives 2 moles of AgCl. We can define a quantity called the gravimetric factor (GF) where:**

Gravimetric Calculations

- Gravimetric Factor (GF):

$$GF = \frac{\text{fwt analyte (g/mol)}}{\text{fwt precipitate (g/mol)}} \times \frac{a(\text{moles analyte})}{b(\text{moles precipitate})}$$

$$GF = \frac{\text{g analyte}}{\text{g precipitate}}$$

$$\%\,(w/w)\text{ analyte (g)} = \frac{\text{wt ppt (g) x GF}}{\text{wt sample (g)}}\ \text{x}\,100\%$$

$$\%\text{ analyte} = \frac{\text{weight analyte (g)}}{\text{weight sample (g)}}\ \text{x}\,100\%$$

$$\%\,(w/w)\text{ analyte (g)} = \frac{\text{wt ppt (g) x GF}}{\text{wt sample (g)}}\ \text{x}\,100\%$$

Gravimetric Errors

- Unknown Stoichiometry:
- Consider Cl^- determination with $AgNO_3$
- $Ag^+ + Cl^- \rightarrow AgCl$
- $Ag^+ + 2\ Cl^- \rightarrow AgCl_2$
- <u>Gravimetric Factor:</u>

$$GF = \frac{\text{fwt analyte}}{\text{fwt precipitate}} \ \text{x}\ \frac{\text{moles analyte}}{\text{moles precipitate}}$$

- <u>Calculation for Cl^-</u> = wt. ppt * GF

- **For example, if you were looking for SO_3 and your precipitate was $BaSO_4$, the gravimetric factor would be:**

	SO_3	**$BaSO_4$**
	analyte	**precipitate**
stoicheometry	**a = 1**	**b = 1**

$$\text{gravimetric factor} = \frac{SO_3}{BaSO_4} = \frac{80.064}{233.391} = 0.3430466$$

The numbers, 80.064 and 233.391, are the formula weights of SO_3 and $BaSO_4$, respectively.

$$\text{gravimetric factor} = \frac{SO_3}{BaSO_4}$$

- **There is only one sulfur in each term and the sulfurs are balanced. In other words, the mole ratio is 1.**

- **Consider the following GF:**
- $$\text{gravimetric factor} = \frac{Ag_2O \text{ (sought)}}{AgCl \text{ (known)}}$$

- **The common element is silver, Ag.**

- $$\text{gravimetric factor} = \frac{Ag_2O}{AgCl}$$

The numbers, 231.7385 and 143.321, are the formula weights of Ag_2O and AgCl, respectively.

- **However, there are two silver atoms represented in the upper term and only one in the lower term.**

$$\text{gravimetric factor} = \frac{Ag_2O}{AgCl}$$

- **To "balance" the silver atoms, a 2 is placed in front of the substance in the lower term.**

$$\text{gravimetric factor} = \frac{Ag_2O}{2\ AgCl}$$

- **The calculation set-up for this gravimetric factor would be:**

$$\text{gravimetric factor} = \frac{Ag_2O}{2\,AgCl} = \frac{231.7385}{2\,(143.321)} = 0.8084502$$

- **Example**
- **Calculate the mg analyte to mg precipitate for the following: P (at wt =30.97) in Ag_3PO_4 (FW = 711.22),**
- **Bi_2S_3 (FW 514.15) in $BaSO_4$ (FW = 233.40)**

- **Answer**

- **P ⇆ Ag_3PO_4**
- **mmol P = mmol Ag_3PO_4**
- **Mg P/30.97 = mg Ag_3PO_4/711.22**
- **Mg P/mg Ag_3PO_4 = 30.97/711.22 = 0.04354**

- **$Bi_2S_3 \leftrightarrows 3\ BaSO_4$**
- mmol Bi_2S_3 = **1/3** mmol $BaSO_4$
- mg Bi_2S_3/FW Bi_2S_3 = **1/3** mg $BaSO_4$/FW $BaSO_4$
- mg Bi_2S_3/514.15 = **1/3** mg $BaSO_4$/233.40
- mg Bi_2S_3/ $BaSO_4$ = **1/3** (514.15/ 233.40) = 0.73429

- Example:- **Phosphate in a 0.2711 g sample was precipitated giving 1.1682 g of $(NH_4)_2PO_4.12\ MoO_3$ (FW = 1876.5). Find percentage P (at wt = 30.97) and percentage P_2O_5 (FW = 141.95) in the sample.**

Answer

- First we set the mol relationship between analyte and precipitate

  - **$P \leftrightarrows (NH_4)_2PO_4.12\ MoO_3$**

- **mmol P = mmol $(NH_4)_2PO_4.12\ MoO_3$**

- **mg P/at wt P = mg $(NH_4)_2PO_4.12\ MoO_3$/ FW $(NH_4)_2PO_4.12\ MoO_3$**
- **mg P = at wt P x (mg $(NH_4)_2PO_4.12\ MoO_3$/ FW $(NH_4)_2PO_4.12\ MoO_3$)**

- **mg P = 30.97 (1.1682×10^3 / 1876.5) = 19.280 mg**
- **% P = (19.280/271.1) x 100 = 7.111%**

The same procedure is applied for finding the percentage of P_2O_5

$$P_2O_5 \leftrightarrows 2\,(NH_4)_2PO_4.12\,MoO_3$$

mmol $P_2O_5 = 1/2\ (NH_4)_2PO_4.12\ MoO_3$

mg $P_2O_5/FW\ P_2O_5 = 1/2\ (mg\ (NH_4)_2PO_4.12\ MoO_3/\ FW\ (NH_4)_2PO_4.12\ MoO_3)$

mg $P_2O_5 = 1/2 \times FW\ P_2O_5 \times (mg\ (NH_4)_2PO_4.12\ MoO_3/\ FW\ (NH_4)_2PO_4.12\ MoO_3)$

mg $P_2O_5 = 1/2 \times 141.95\ (1.1682\times10^3/1876.5) = 44.185$ **mg**

$\%\ P_2O_5 = (44.185/271.1) \times 100 = 16.30\%$

Weight of substance sought = weight of precipitate x GF

One can also consider the problem by looking at the number of mmoles of analyte in terms of the mmoles of the precipitate where for the precipitation of Cl_2 as AgCl, we can write

$$\mathbf{Cl_2 \leftrightarrows 2\ AgCl}$$

$$\mathbf{mmol\ Cl_2 = 1/2\ mmol\ AgCl}$$

$$\mathbf{(mg\ Cl_2/FW\ Cl_2) = 1/2\ (mg\ AgCl/FW\ AgCl)}$$

Manganese in a 1.52 g sample was precipitated as Mn_3O_4 (FW = 228.8) weighing 0.126 g. Find percentage Mn_2O_3 (FW = 157.9) and Mn (at wt = 54.94) in the sample.

Solution

$$3\ Mn_2O_3 \leftrightarrows 2\ Mn_3O_4$$

$\text{mmol } Mn_2O_3 = 3/2 \text{ mmol } Mn_3O_4$

$\text{mg } Mn_2O_3 / \text{FW } Mn_2O_3 = 3/2 (\text{mg } Mn_3O_4/\text{FW } Mn_3O_4)$

$\text{mg } Mn_2O_3 = 3/2 \text{ FW } Mn_2O_3 (\text{mg } Mn_3O_4/\text{FW } Mn_3O_4)$

$\text{mg } Mn_2O_3 = 3/2 \times 157.9 (126/228.8) = 130 \text{ mg}$

$\% \ Mn_2O_3 = (130/1520) \times 100 = 8.58\%$

The same idea is applied for the determination of Mn in the sample

$$3\ Mn \leftrightarrows Mn_3O_4$$

mmol Mn = 3 mmol Mn_3O_4

mg Mn / at wt Mn = 3 (mg Mn_3O_4/FW Mn_3O_4)

mg Mn = 3 at wt Mn (mg Mn_3O_4/FW Mn_3O_4)

mg Mn = 3 x 54.94 (126/228.8) = 90.8 mg

% Mn = (90.8/1520) x 100 = 5.97%

What weight of sulfur (FW = 32.064) ore which should be taken so that the weight of $BaSO_4$ (FW = 233.40) precipitate will be equal to half of the percentage sulfur in the sample.

Solution

$$S \leftrightarrows BaSO_4$$

mmol S = mmol $BaSO_4$

mg S/at wt S = mg $BaSO_4$/FW $BaSO_4$

mg S = at wt S x (mg $BaSO_4$ / FW $BaSO_4$)

mg S = 32.064 x (1/2 %S/233.40)

mg S = 0.068689 %S

%S = (mg S/mg sample) x 100

%S = 0.068689 %S/mg sample) x 100

mg sample = 0.068689 %S x 100 /%S = 6.869 mg

A mixture containing only $FeCl_3$ (FW = 162.2) and $AlCl_3$ (FW = 133.34) weighs 5.95 g. The chlorides are converted to hydroxides and ignited to Fe_2O_3 (FW = 159.7) and Al_2O_3 (FW = 101.96). The oxide mixture weighs 2.62 g. Calculate the percentage Fe (at wt = 55.85) and Al (at wt = 26.98) in the sample.

Solution

g $FeCl_3$ + g $AlCl_3$ = 5.95

g Fe_2O_3 + g Al_2O_3 = 2.62

$Fe \leftrightarrows FeCl_3$

1 mol Fe = 1 mol $FeCl_3$

g Fe/at wt Fe = g $FeCl_3$/ FW $FeCl_3$

Rearrangement gives

g $FeCl_3$ = g Fe (FW $FeCl_3$/at wt Fe)

In the same manner

g $AlCl_3$ = g Al (FW $AlCl_3$/at wt Al)

g $FeCl_3$ + g $AlCl_3$ = 5.95

g Fe (FW $FeCl_3$/at wt Fe) + g Al (FW $AlCl_3$/at wt Al) = 5.95

assume g Fe = *x*, g Al = y then:

x (FW $FeCl_3$/at wt Fe) + y (FW $AlCl_3$/at wt Al) = 5.95

x (162.2/55.85) + y (133.34/26.98) = 5.95

2.90 x + 4.94 y = 5.95 **(1)**

The same treatment with the oxides gives

$$2\,Fe \leftrightarrows Fe_2O_3$$

mol Fe = 2 mol Fe_2O_3

g Fe/at wt Fe = 2 (g Fe_2O_3/FW Fe_2O_3)

g Fe_2O_3 = 1/2 g Fe (FW Fe_2O_3/at wt Fe)

In the same manner

g Al_2O_3 = 1/2 g Al (FW Al_2O_3/at wt Al)

g Fe_2O_3 + g Al_2O_3 = 2.62

1/2 g Fe (FW Fe_2O_3/at wt Fe) + 1/2 g Al (FW Al_2O_3/at wt Al) = 2.62

$$1/2\,x\,(159.7/55.85) + 1/2\,y\,(101.96/26.98) = 2.62$$

$$1.43\,x + 1.89\,y = 2.62 \qquad (2)$$

$2.90\,x + 4.94\,y = 5.95$ (1)

$1.43\,x + 1.89\,y = 2.62$ (2)

from (1) and (2) we get

$x = 1.07$

$y = 0.58$

% Fe = (1.07/5.95) x 100 = 18.0%

% Al = (0.58/5.95) x 100 = 9.8%

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Problem

Consider a 1.0000 g sample containing 75% potassium sulfate (FW 174.25) and 25% MSO_4. The sample is dissolved and the sulfate is precipated as $BaSO_4$ (FW 233.39). If the $BaSO_4$ ppt weighs 1.4900g, what is the atomic weight of M^{2+} in MSO_4?

$$K_2SO_4 \rightarrow BaSO_4$$

$$MSO_4 \rightarrow BaSO_4$$

mmol $BaSO_4$ = mmol K_2SO_4 + mmol MSO_4

$$\frac{1.4900}{233.39} = \frac{0.75}{174.25} + \frac{0.25}{x+96}$$

Rearranging and solving:

$$0.4855 = \frac{58.3475}{x+96.06}; x = 24.12$$

A mixture of mercurous chloride (FW 472.09) and mercurous bromide (FW 560.99) weighs 2.00 g. The mixture is quantitatively reduced to mercury metal (At wt 200.59) which weighs 1.50 g. Calculate the % mercurous chloride and mercurous bromide in the original mixture.

$Hg_2Cl_2 \rightarrow 2Hg$

$Hg_2Br_2 \rightarrow 2Hg$

Answer

mmol Hg = 2 mmol Hg_2Cl_2 + 2 mmol Hg_2Br_2

$$\frac{1.50}{200.59} = \frac{2 \times x}{472.09} + \frac{2 \times (2 - x)}{560.99}$$

Rearranging and solving:

$$0.8498 + 0.7151(2 - x) = 1.50$$

$$x = 0.5182 g$$

A 10.0 mL solution containing Cl^- was treated with excess $AgNO_3$ to precipitate 0.4368 g of AgCl. What was the concentration of Cl^- in the unknown? (AgCl = 143.321 g/mol)

mmol Cl^- = mmol AgCl

$$= \frac{436.8\,\text{mg}}{143.321} = 3.048\,\text{mmol}$$

Concentration of Cl^- $= \dfrac{3.048\,\text{mmol}}{10\,\text{mL}} = 0.3048\,\text{M}$

Calculating Results from Gravimetric Data

- The calcium in a 200.0 mL sample of a natural water was determined by precipitating the cation as CaC_2O_4. The precipitate was filtered, washed, and ignited in a crucible with an empty mass of 26.6002 g. The mass of the crucible plus CaO (fwt 56.077 g/mol) was 26.7134 g. Calculate the concentration of Ca (fwt 40.078 g/mol) in the water in units of grams per 100 mL.

Calculating Results from Gravimetric Data

- An iron ore was analyzed by dissolving a 1.1324 g sample in concentrated HCl. The resulting solution was diluted with water, and the iron(III) was precipitated as the hydrous oxide $Fe_2O_3 \cdot xH_2O$ by addition of NH_3. After filtration and washing, the residue was ignited at high temperature to give 0.5394 g pure Fe_2O_3 (fwt 159.69 g/mol). Calculate (a) the % Fe (fwt 55.847 g/mol) and (b) % Fe_3O_4 (fwt 231.54 g/mol) in the sample.

Calculating Results from Gravimetric Data

- A 0.2356 g sample containing only NaCl (fwt 58.44 g/mol) and $BaCl_2$ (fwt 208.23 g/mol) yielded 0.4637 g of dried AgCl (fwt 143.32 g/mol). Calculate the percent of each halogen compound in the sample.

Precipitation Equilibria: The Solubility Product

- Solubility of Slightly Soluble Salts:
- $AgCl_{(s)} \leftrightarrow (AgCl)_{(aq)} \leftrightarrow Ag^+ + Cl^-$
- Solubility Product K_{SP} = ion product
- $K_{SP} = [Ag^+][Cl^-]$
- $Ag_2CrO_{4(s)} \leftrightarrow 2\ Ag^+ + CrO_4^{2-}$
- $K_{SP} = [Ag^+]^2[CrO_4^{2-}]$

Precipitation Equilibria: The Common Ion Effect

- Common Ion Effect
- Will decrease the solubility of a slightly soluble salt.

Calculating Results from Gravimetric Data

- The calcium in a 200.0 mL sample of a natural water was determined by precipitating the cation as CaC_2O_4. The precipitate was filtered, washed, and ignited in a crucible with an empty mass of 26.6002 g. The mass of the crucible plus CaO (fwt 56.077 g/mol) was 26.7134 g. Calculate the concentration of Ca (fwt 40.078 g/mol) in the water in units of grams per 100 mL.

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- Calculate the weight of Ba and the weight of Cl present in 25.0 g $BaCl_2$.
- (a)

$$\frac{gCl}{gBaCl_2} = \frac{at.wt\ of\ Cl}{f.wt\ of\ BaCl_2} \times \frac{2}{1}$$

$$gCl = gBaCl_2 \times \frac{35.5}{208} \times \frac{2}{1}$$

$$gCl = 25.0\,x\,\frac{35.5}{208}\,x\,\frac{2}{1}$$

$$gCl = 8.5\,g$$

$$\frac{gCl}{gAgCl} = \frac{at.t.Cl}{f.wt.of\,Agcl} \text{ x } \frac{1}{1}$$

$$\frac{gCl_2}{gAgCl} = \frac{f.wt.Cl_2}{f.wt.Agcl} \text{ x } \frac{1molCl_2}{2molAgCl}$$

- Q). Al in an ore is determined by dissolving it and then participating with base as $Al(OH)_3$ and igniting to Al_2O_3, which is weighed. What weight of Al was I the sample if the ignited precipitate weighed 0.2385 g?
-

- Q). The mass of CaO = 26.7134 –26.6002 g = 0.1132 g.
- g Ca = ?

$$\frac{gCa}{gCaO} = \frac{at.wt\ of\ .Ca}{f.wt.ofCaO} x \frac{1}{1}$$

$$gCa = gCaO\ x\ \frac{40}{56} x \frac{1}{1}$$

$$gCa = 0.1132g\ x\ \frac{40}{56}$$

$$gCa = 2.01x10^{-3}\ \cancel{mol}$$

$$con.Ca(g/100mL) = \frac{2.018x10^{-3}\ \cancel{mol}\ x\ 40g/\cancel{mol}}{200\,mL} x100\,mL$$

$$= 0.04\,g/100\,mL$$

- Q). A 0.2356 g sample containing only NaCl (58.44 g/mol) and $BaCl_2$ (208 g/mol) yielded 0.4637 g of dried AgCl(143.32 g/mol). Calculate the percentage of each halogen compound in the sample.

- (ans):-
- x +y = 0.2356 g.
- Where x = mass of NaCl and y is the mass of $BaCl_2$.
- Amount of

$$\frac{gAgCl}{gNaCl} = \frac{f.wtofAgCl}{f.wt.ofNaCl} x \frac{1}{1}$$

$$gAgCl = gNaCl \ \ x \frac{143}{58} x \frac{1}{1}$$

$$gAgCl = 2.45 x\ gAgCl$$

similarly

$$\frac{gAgCl}{gBaCl_2} = \frac{f.wtof\ AgCl}{f.wt.of\ BaCl_2} x \frac{2}{1}$$

$$gAgCl = gBaCl_2\ x \frac{143}{58} x \frac{1}{1}$$

$$gAgCl = 1.38y\, g\ BaCl_2$$

Precipitation Equilibria

- Inorganic solids which have limited water solubility show an equilibrium in solution represented by the so called solubility product. For example, AgCl slightly dissolve in water giving Cl^- and Ag^+ where
-
- **$AgCl_{(s)} \leftrightarrows Ag^+ + Cl^-$**
- **$K = [Ag^+][Cl^-]/[AgCl_{(s)}]$**
-
- However, the concentration of a solid is constant and the equilibrium constant can include the concentration of the solid and thus is referred to, in this case, as the solubility product, k_{sp} where
-
- **$K_{sp} = [Ag^+][Cl^-]$**
-

- It should be clear that the product of the ions raised to appropriate power as the number of moles will fit in one of three cases:
- 1. **When the product is less than K_{sp}**: No precipitate is formed and we have a clear solution
- 2. **When the product is equal to k_{sp}** : We have a saturated solution
- 3. **When the product exceeds the k_{sp}** : A precipitate will form
-
- It should also be clear that at equilibrium of the solid with its ions, the concentration of each ion is constant and the precipitation of ions in solution does occur but at the same rate as the solubility of precipitate in solution. Therefore, the concentration of the ions remains constant at equilibrium.

- Three situations will be studied:
- **a. Solubility in pure water**
- **b. Solubility in presence of a common ion**
- **c. Solubility in presence of diverse ions**

- **Two other situations will be discussed in later chapters**
- **Solubility in acidic solutions**
- **solubility in presence of a complexing agent**

Solubility in Pure Water

Example

Calculate the concentration of Ag^+ and Cl^- in pure water containing solid AgCl if the solubility product is 1.0×10^{-10}.

Solution

First, we set the stoichiometric equation and assume that the molar solubility of AgCl is s.

$$\mathbf{AgCl_{(s)} \leftrightarrows Ag^+ + Cl^-}$$

Before Equilibrium	**Solid**	**0**	**0**
Equation	**$AgCl_{(s)}$**	**Ag^+**	**Cl^-**
At Equilibrium	**Solid**	**s**	**s**

$K_{sp} = [Ag^+][Cl^-]$

$1.0x10^{-10} = s \text{ x } s = s^2$

$s = 1.0 \text{ x } 10^{-5} \text{ M}$

$[Ag^+] = [Cl^-] = 1.0 \text{ x } 10^{-5} \text{ M}$

Example

Calculate the molar solubility of $PbSO_4$ in pure water if the solubility product is 1.6×10^{-8}

Solution

Before Equilibrium	**Solid**	**0**	**0**
Equation	$\mathbf{PbSO_{4(s)}}$	$\mathbf{Pb^{2+}}$	$\mathbf{SO_4{}^{2-}}$
At Equilibrium	**Solid**	**s**	**s**

$\mathbf{K_{sp} = [Pb^{2+}][SO_4{}^{2-}]}$

$\mathbf{1.6 \times 10^{-8} = s \times s = s^2}$

$\mathbf{s = 1.3 \times 10^{-4}\ M}$

Calculate the molar solubility of PbI_2 in pure water if the solubility product is 7.1×10^{-9}.

Solution

Before Equilibrium	**Solid**	**0**	**0**
Equation	$\mathbf{PbI_{2(s)}}$	$\mathbf{Pb^{2+}}$	$\mathbf{2I^-}$
At Equilibrium	**Solid**	**s**	**2s**

$K_{sp} = [Pb^{2+}][I^-]^2$

$7.1\text{x}10^{-9} = s \text{ x } (2s)^2$

$7.1\text{x}10^{-9} = 4s^3$

$s = 1.2\text{x}10^{-3}$ M

- If we compare the molar solubilities of $PbSO_4$ and PbI_2 we find that the solubility of lead iodide is larger than that of lead sulfate although lead iodide has smaller k_{sp}.
- You should calculate solubilities rather than comparing solubility products to check which substance gives a higher solubility.
- In presence of a precipitating agent, the substance with the least solubility will be precipitated first.

- **What must be the concentration of Ag^+ to just start precipitation of AgCl in a $1.0x10^{-3}$ M NaCl solution.**
-
- **Solution**
-
- **AgCl just starts to precipitate when the ion product just exceeds k_{sp}**
- **$K_{sp} = [Ag^+][Cl^-]$**
- **$1.0x10^{-10} = [Ag^+] \times 1.0 \times 10^{-3}$**
- **$[Ag^+] = 1.0x10^{-7}$ M**

- **What pH is required to just start precipitation of Fe(III) hydroxide from a 0.1 M $FeCl_3$ solution. $K_{sp} = 4x10^{-38}$.**

- **Solution**

- **$Fe(OH)_3 \leftrightarrows Fe^{3+} + 3\ OH^-$**
- **$K_{sp} = [Fe^{3+}][OH^-]^3$**
- **$4x10^{-3} = 0.1\ x\ [OH^-]^3$**
- **$[OH^-] = 7x10^{-13}$ M**
- **pH = 14 – pOH**
- **pH = 14 – 12.2 = 1.8**

Solubility in Presence of a Common Ion

- **10 mL of 0.2 M $AgNO_3$ is added to 10 mL of 0.1 M NaCl. Find the concentration of all ions in solution and the solubility of AgCl formed.**

- **First we find mmol $AgNO_3$ and mmol NaCl then determine the excess concentration**
-
- **mmol Ag^+ = 0.2 x 10 = 2**
- **mmol Cl^- = 0.1 x 10 = 1**
- **mmol AgCl formed = 1 mmol**
- **mmol Ag^+ excess = 2 – 1 = 1**
- **$[Ag^+]_{excess}$ = mmol/mL = 1/20 = 0.05 M**

- **We can find the concentrations of NO_3^- = 0.2 x 10/20 = 0.1 M**
- **$[Na^+]$ = 0.1*10/20 = 0.05 M**
- **Chloride ions react to form AgCl, therefore the only source for Cl^- is the solubility of AgCl; but now in presence of 0.05 M excess Ag^+**

Before Equilibrium	**Solid**	**0.05**	**0**
Equation	**$AgCl_{(s)}$**	**Ag^+**	**Cl^-**
At Equilibrium	**Solid**	**0.05 + s**	**s**

- $K_{sp} = [Ag^+][Cl^-]$
- $1.0 \times 10^{-10} = (0.05 + s)\ (s)$
- Since the solubility product is very small, we can assume $0.05 \gg s$
- $1.0 \times 10^{-10} = 0.05\ (s)$
- $s = 2.0 \times 10^{-9}$ M
- Relative error = $(2.0 \times 10^{-9} / 0.05) \times 100 = 4 \times 10^{-6}\%$ which is extremely small, therefore:
- $[Cl^-] = 2.0 \times 10^{-9}$ M
- $[Ag^+] = 0.05 + 2.0 \times 10^{-9} = 0.05$ M
-
- Look at how the solubility is decreased in presence of the common ion.

- **25 mL of 0.100 M $AgNO_3$ are mixed with 35 mL of 0.050 M K_2CrO_4. Find the concentration of each ion in solution at equilibrium if k_{sp} of $Ag_2CrO_4 = 1.1 \times 10^{-12}$.**
-
- **Solution**
-
- **$2\ Ag^{2+} + CrO_4^{2-} \rightarrow Ag_2CrO_{4(s)}$**
-
- **mmol $Ag^+ = 0.100 \times 25 = 2.5$**
- **mmol $CrO_4^{2-} = 0.050 \times 35 = 1.75$**
- **From the stoichiometry we know that two moles of silver react with one mole of chromate, and chromate will be in excess, therefore:**

- The concentration of chromate left in as follows:

- mmol CrO_4^{2-} left = initial mmol CrO_4^{2-} - mmol CrO_4^{2-} reacted
- mmol CrO_4^{2-} left = initial mmol CrO_4^{2-} - ½ mmol Ag^+
- mmol CrO_4^{2-} left = 0.05 x 35 – ½ x 0.1 x 25 = 0.5

- **mmol CrO_4^{2-} excess = 1.75 – 1.25 = 0.50**
- **[CrO_4^{2-}] excess = mmol/mL = 0.5/60 = 0.0083 M**
- **[NO_3^-] = mmol/mL = 0.100 x 25/60 = 0.0417 M**
- **[K^+] = mmol/mL = 2 x 0.050 x 35/60 = 0.0583 M**
- **The silver was consumed in the reaction and the only way to calculate its concentration is through the solubility product:**

Before Equilibrium	Solid	0	0.0083
Equation	$Ag_2CrO_{4(s)}$	$2Ag^+$	CrO_4^{2-}
At Equilibrium	Solid	2s	0.0083 + s

- $K_{sp} = [Ag^+]^2 [CrO_4^{2-}]$
- $1.1x10^{-12} = (2s)^2 * (0.0083 + s)$
- However, s is very small since the equilibrium constant is very small. Therefore, assume 0.0083>>s
- $1.1x10^{-12} = (2s)^2 \text{ x } 0.0083$
- $s = 5.7_6 x10^{-6}$ M

- Relative error = $(5.7_6 x10^{-6}/0.0083)$ x 100 = 0.07%
- $[Ag^+] = 2s = 2x5.7_6 x10^{-6} = 1.1_5 x10^{-5}$ M
- $[CrO_4^{2-}] = 0.0083 + s = 0.0083 + 5.7_6 x10^{-6} = 0.0083$ M
-

▪ Solubility in Presence of Diverse Ions

- **As expected from previous information, diverse ions have a screening effect on dissociated ions which leads to extra dissociation.**
- **Solubility will show a clear increase in presence of diverse ions as the solubility product will increase. Look at the following example:**

- **Example**

- **Find the solubility of AgCl (k_{sp} = 1.0 x 10^{-10}) in 0.1 M $NaNO_3$. The activity coefficients for silver and chloride are 0.75 and 0.76, respectively.**

- **Answer**
- **$AgCl_{(s)} \leftrightarrows Ag^+ + Cl^-$**

- We use activities instead of concentrations since the solution contains diverse ions where we have:
- $K_{sp,\ Th} = a_{Ag^+}\, a_{Cl^-}$
- $K_{sp,\ Th} = [Ag^+]\, f_{Ag^+} * [Cl^-]\, f_{Cl^-}$
- $1.0x10^{-10} = s \text{ x } 0.75 \text{ x } s \text{ x } 0.76$
- $s = 1.3x10^{-5}$ M

- **The solubility of AgCl in pure water can be calculated to be 1.0×10^{-5} M. If we compare this value to that obtained in presence of diverse ions we see**

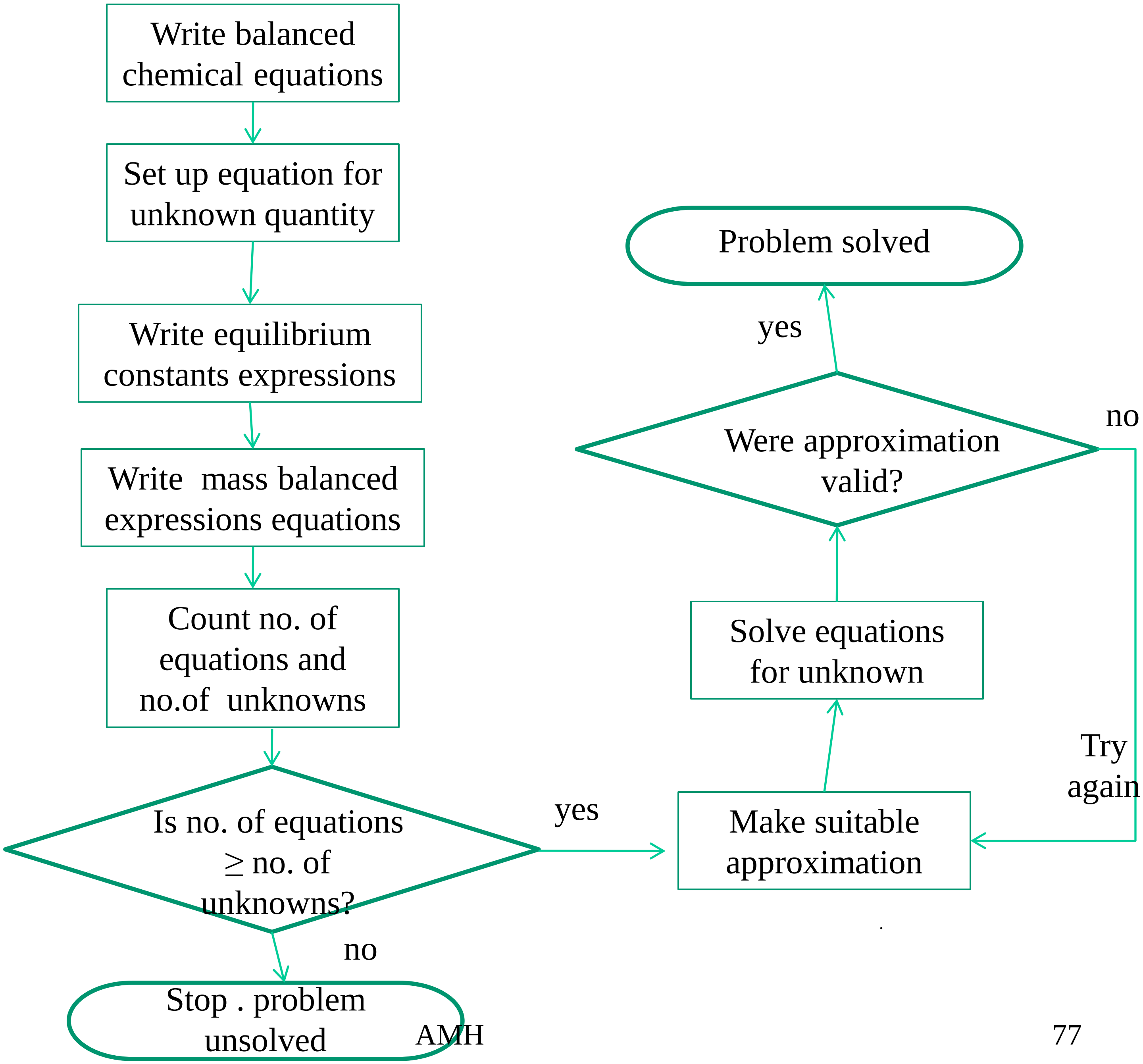
- **%increase in solubility = $\{(1.3 \times 10^{-5} - 1.0 \times 10^{-5})/1.0 \times 10^{-5}\} \times 100 = 30\%$**

- **Therefore, once again we have an evidence for an increase in dissociation or a shift of equilibrium to right in presence of diverse ions.**

Systematic Approach to Equilibrium Calculations

- How to Solve Any Equilibrium Problem
- 1. Write balanced chemical reactions
- 2. Write equilibrium constant expressions
- 3. Write all mass balance expressions
- 4. Write the charge balance expression
- 5. Equations >= Chemical Species sol possible
- 6. Make assumptions where possible
- 7. Calculate answer
- 8. Check validity of assumptions

- Flow chart for systematic approach to multiple equilibria calculations


Write balanced chemical equations
Set up equation for unknown quantity
Write equilibrium constants expressions
Write mass balanced expressions equations
Count no. of equations and no.of unknowns
Is no. of equations ≥ no. of unknowns?
yes
no
Stop . problem unsolved
Make suitable approximation
Solve equations for unknown
Were approximation valid?
yes
no
Try again
Problem solved

Mass Balance Equations

- In a mass balance, the analytical concentration is equal to the sum of the concentrations of the equilibrium species derived from the parent compound.
- The equations for all the pertinent chemical equilibria are written from which appropriate relations between species concentrations are written.

- Write the equation of mass balance for a 0.100 M solution of acetic acid (HOAc)
- The equilibria are:
- $$HOAc \rightleftharpoons H^+ + OAc^-$$
- $$H_2O \rightleftharpoons H^+ + OH^-$$
- We know that the analytical concentration of HOAc is equal to the sum of the equilibrium concentrations of all its species.
- $C_{HOAc} = [HOAc] + [OAc^-] = 0.100\ M$

- A second mass balance expression may be written for the equilibrium concentration of H^+.
- Which is derived from both HOAc and H_2O.
- $[H^+] = [OAc^-] + [OH^-]$
- Eg (02):-
- Write the equation of mass balance for a 1.00×10^{-5} M , $[Ag(NH_3)_2]Cl$ solution.

- $[Ag(NH_3)_2]Cl \rightarrow Ag(NH_3)^{2+} + Cl^-$
- $Ag(NH_3)^{2+} \rightleftharpoons Ag(NH_3)^+ + NH_3$
- $Ag(NH_3)^+ \rightleftharpoons Ag+ + NH_3$
- $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$
- $H_2O \rightleftharpoons H^+ + OH^-$
- $C_{Ag} = [Ag^+] + [Ag(NH_3)^+] + [Ag(NH_3)^{2+}]$

  $= [Cl^-] = 1.00 \times 10^{-5}$ M
- Further, for N – containing species:
- The mass balance equation is:

- $C_{NH_3} = [NH_4^+] + [NH_3] + [Ag(NH_3)^+] + 2[Ag(NH_3)^{2+}] = 2 \times 1.00 \times 10^{-5}$ M = 2.00×10^{-5} M.
- Eg (03)
- Write mass balance expressions for a 0.0100 M solution of HCl that is in equilibrium with an excess of solid $BaSO_4$.
- $BaSO_4 \rightleftharpoons Ba^{2+} + SO_4^{2-}$
- $SO_4^{2-} + H_2O \rightleftharpoons HSO_4^- + H_2O$
- $2H_2O \rightleftharpoons H_3O^+ + OH^-$

- $[Ba^{2+}] = [SO_4^{2-}] + [HSO_4^-]$
- The H_3O^+ in the solution can exist either as free H_3O^+ ions or combined with SO_4^{2-} to form HSO_4^-.
- H_3O^+ ions have two sources. From HCl and the dissociation of water.
- $[H_3O^+] + [HSO_4^-] = C_{HCl} + [OH^-] = 0.0100 + [OH^-]$

Charge Balance Equation

- There is no solution containing a detectable excess of positive or negative charge.
- The sum of the positive charges equals to sum of the negative charges.
- We may write single charge balance equation for a given set of equilibria.

- Consider for a 0.100 M solution of NaCl.
- $NaCl \rightleftharpoons Na^+ + Cl^-$
- $2H_2O \rightleftharpoons H_3O^+ + OH^-$
- $[Na^+] + [H_3O^+] = [Cl^-] + [OH^-]$

- Write a charge balance equation for a solution of H_2S.
- $H_2S \rightleftharpoons H^+ + HS^-$
- $HS^- \rightleftharpoons H^+ + S^{--}$
- $H_2O \rightleftharpoons H^+ + OH^-$

- For a singly charged species, the charge concentration is identical to the concentration of the species.
- But for S^{--} the charge concentration is twice tat of the species.
- $$[H^+] = 2[S^{--}] + [HS^-] + [OH^-]$$
- Write a charge balance expression for a solution containing KCl,$Al_2(SO_4)_3$, and KNO_3. Neglect the dissociation of water.

- $KCl \rightarrow K^+ + Cl^-$
- $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{--}$
- $KNO_3 \rightarrow K^+ + NO_3^-$
- $[K^+] + [Al^{3+}] = [Cl^-] + [SO_4^{--}] + [NO_3^-]$
- But correctly, the charge balance equation as:
- $[K^+] + 3[Al^{3+}] = [Cl^-] + 2[SO_4^{--}] + [NO_3^-]$

Multiple Chemical Equilibria

- Dissolve $NaHCO_3$ in water:
  - $NaHCO_3 \rightarrow Na^+ + HCO_3^-$
- $HCO_3^- + HOH \leftrightarrow H_2CO_3 + OH^-$ $K_e = Kb = Kw/K_{a1}$
- $HCO_3^- + HOH \leftrightarrow H_3O^+ + CO_3^{2-}$ $K_e = K_{a2}$
- $HOH + HOH \leftrightarrow H_3O^+ + OH^-$ $K_e = K_w$
- 5 chemical species affected by 3 equilibria
- Equilibrium constants do NOT change with chemical additions/ deletions
  - Add Ba^{2+}: $Ba^{2+} + CO_3^{2-} \leftrightarrow BaCO_{3(s)}$ $K_e = K_{sp}$
  - (Now there are 6 species affected by 4 equilibria)
- **Note: For Polyprotic Acids (H_NA): $K_{(step)} = K_{a1}, K_{a2}$,-- K_{aN}**
- $H_2A + HOH \leftrightarrow H_3O^+ + HA^-$ $K_e = K_{a1}$
- $HA^- + HOH \leftrightarrow H_3O^+ + A^{2-}$ $K_e = K_{a2}$

Multiple Chemical Equilibria Compositional Calculations

- **What are concentrations of individual species?**
- For N species, M equilibria
- Need, N independent algebraic expressions:
- Equilibrium expressions ($M < N$)
- Mass balance statements
- Charge Balance statements

Multiple Chemical Equilibria Compositional Calculations

- **Mass Balance Equations**:
- Relate equilibrium concentrations of species
  - Stoichiometric relationships
  - How the solution was prepared
  - What kinds of equilibria exist
- E.g. 0.1 M HNO_2 (C_{HA} = 0.1 M)
- $HNO_2 + HOH \leftrightarrow H_3O^+ + NO_2^-$ $\quad K_e = K_a$
- C_{HA} $\qquad$ (x) $\qquad$ (x)
- $HOH + HOH \leftrightarrow H_3O^+ + OH^-$ $\quad K_e = K_w$
- $\qquad$ (w) $\qquad$ (w)
- $C_{HA} = [HNO_2] + [NO_2^-]$ $\qquad$ all forms of "HNO_2"
- $[H_3O^+] = [OH^-] + [NO_2^-] = w + x$ $\quad$ $[H_3O^+]$ from 2 sources

Multiple Chemical Equilibria Compositional Calculations

- **<u>Charge Balance Equations:</u>**
- In any electrolyte solution,
- amt. of <u>positive</u> charge = amt. of <u>negative</u> charge
- solution charge for each species = [conc.][charge/ion]
- E.g. for solution $MgCO_3$
- $MgCO_3 \rightarrow Mg^{2+} + CO_3^{2-}$ leads to equilibria:
- $CO_3^{2-} + HOH \leftrightarrow HCO_3^- + OH^-$
- $HCO_3^- + HOH \leftrightarrow H_2CO_3 + OH^-$
- **<u>Charge Balance</u>**:
- $2[Mg^{2+}] + [H_3O^+] = 2[CO_3^{2-}] + [HCO_3^-] + [OH^-]$

Multiple Chemical Equilibria Example Problem (1)

- **<u>What is pH of $Mg(OH)_2$ solution ?</u> (assume $a_x = [X]$)**
- **<u>Equilibria:</u>**
- **$Mg(OH)_{2(s)} \leftrightarrow Mg^{2+} + 2\ OH^-$ $K_{sp} = [Mg^{2+}][OH^-] = 1.8 \times 10^{-11}$**
- $HOH + HOH \leftrightarrow H_3O^+ + OH^-$ $K_e = K_w = 1.0 \times 10^{-14}$
- **<u>Mass Balance:</u>** $[OH^-] = [H_3O^+] + 2\ [Mg^{2+}]$
- **<u>Charge Balance:</u>** $[OH^-] = [H_3O^+] + 2\ [Mg^{2+}]$
- **<u>Expression for unknown :</u>** $[H_3O^+] = K_w / [OH^-]$

Multiple Chemical Equilibria Example Problem (1)

- **What is pH of $Mg(OH)_2$ solution ? (assume a_x= [X])**
- **Expression for unknown :** $[H_3O^+] = K_w / [OH^-]$
- **Solution:** Assume $[H_3O^+] << 2\,[Mg^{2+}]$; $[OH^-] = 2\,[Mg^{2+}]$
- Substitute into K_{sp}, $K_{sp} = ([OH^-]/2)([OH^-])^2$
- $[OH^-]^3 = 2\,K_{sp}$; $[OH^-] = (2\,K_{sp})^{1/3} = 3.3 \times 10^{-4}$ M
- $[H_3O^+] = K_w/(3.3 \times 10^{-4}) = 3.0 \times 10^{-11}$ M
- pH = $-\log [H_3O^+] = 10.52$ (2 sig figs)
- Note: original approximation was OK!
- i.e. $[H_3O^+] << [OH^-]$ ($3.0 \times 10^{-11} << 3.3 \times 10^{-4}$)
- Assumed $[OH^-] = 2\,[Mg^{2+}]$; $[H_3O^+] << 2\,[Mg^{2+}]$
- Assumed $[H_3O^+] << [OH^-]$ ➔ **OK**!

Multiple Chemical Equilibria Example Problem (2)

- **What is pH of 0.1 M Na_3PO_4 solution?** **(C_s = 0.1 M)**
- $Na_3PO_{4(s)} \rightarrow 3Na^+ + PO_4^{3-}$
- **Equilibria:**
- (1) $PO_4^{3-} + HOH \leftrightarrow HPO_4^{2-} + OH^-$ $K_w/K_{a3} = 2.38 \times 10^{-2}$
- (2) $HPO_4^{2-} + HOH \leftrightarrow H_2PO_4^- + OH^-$ $K_w/K_{a2} = 1.6 \times 10^{-9}$
- **(3)** $H_2PO_4^- + HOH \leftrightarrow H_3PO_4 + OH^-$ $K_w/K_{a1} = 1.4 \times 10^{-12}$
- **(4)** $HOH + HOH \leftrightarrow H_3O^+ + OH^-$ $K_w = 1.0 \times 10^{-14}$
- **Mass Balance Equations:**
- (5) $C_s = [PO_4^{3-}] + [HPO_4^{2-}] + [H_2PO_4^-] + [H_3PO_4]$
- (6) $[OH^-] = [H_3O^+] + [HPO_4^{2-}] + [H_2PO_4^-] + [H_3PO_4]$
- (*) $K_{a1} = 7.1 \times 10^{-3}$ $K_{a2} = 6.3 \times 10^{-8}$ $K_{a3} = 4.2 \times 10^{-13}$

Multiple Chemical Equilibria Example Problem (2)

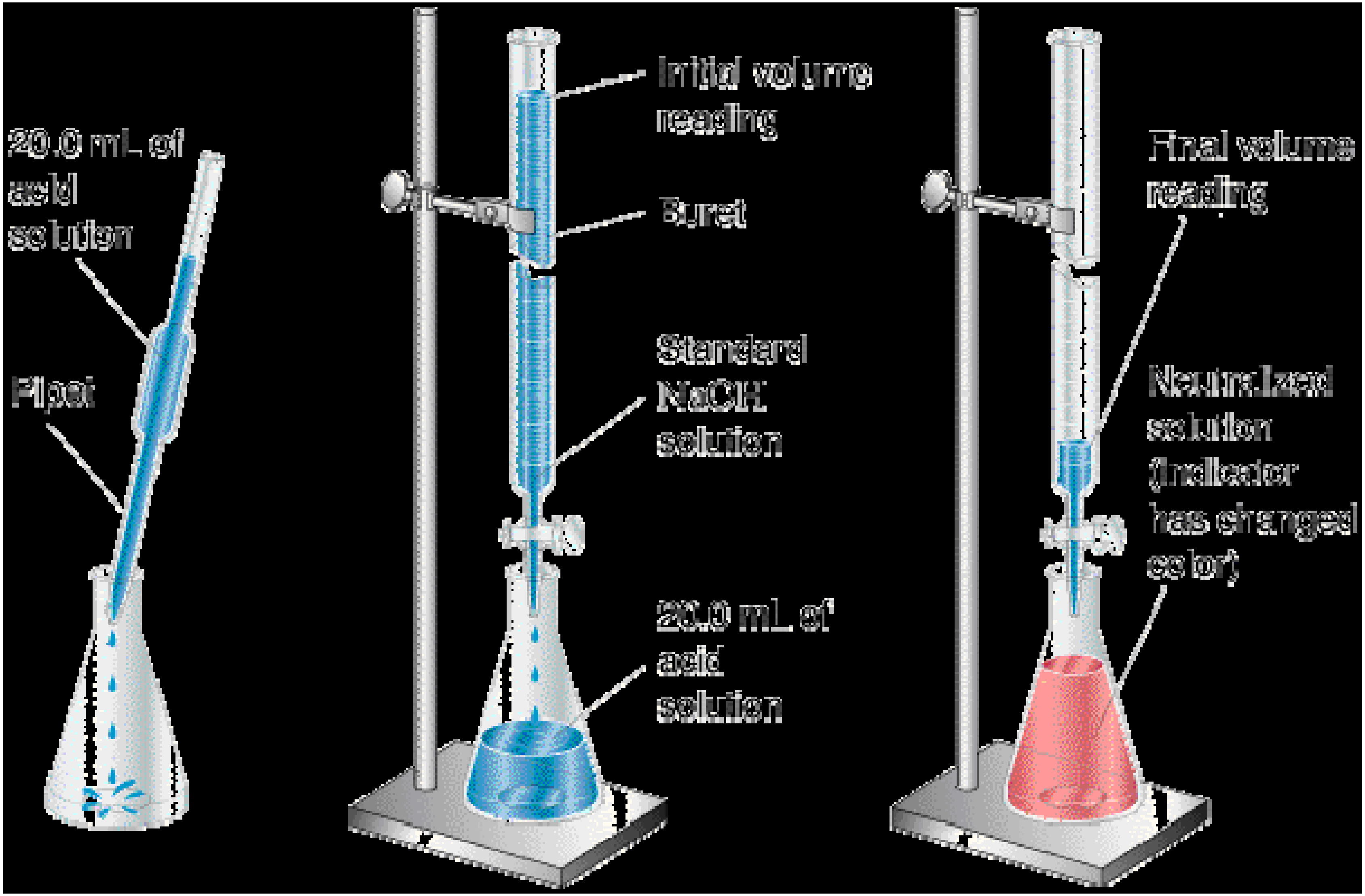
- **<u>What is pH of 0.1 M Na_3PO_4 solution?</u>** **(C_s = 0.1 M)**
- **<u>Charge Balance:</u>**
- (7) $[H_3O^+] + [Na^+] = [OH^-] + [H_2PO_4^-] + 2\,[HPO_4^{2-}] + 3\,[PO_4^{3-}]$
- Note: $[Na^+] = 3\,C_s$
- **<u>Is problem solvable?</u>**
- 6 unknowns $[H_3O^+]$,$[OH^-]$,$[H_2PO_4^-]$,$[HPO_4^{2-}]$,$[PO_4^{3-}]$, $[H_3PO_4]$
- 7 equations (see 1-7)
- Thus, the problem should be solvable.

Multiple Chemical Equilibria Example Problem (2)

- **What is pH of 0.1 M Na_3PO_4 solution?** **(C_s = 0.1 M)**
- **Assume** $[H_3O^+] + [H_2PO_4^-] + [H_3PO_4] << [HPO_4^{2-}]$
- Then from equation (6): $[OH^-] = [HPO_4^{2-}] = x$
- Also Assume $[HPO_4^{2-}] + [H_2PO_4^-] + [H_3PO_4] << [PO_4^{3-}]$
- Then from equation (5): $[PO_4^{3-}] = C_s = 0.1$ M
- equation (1): $[HPO_4^{2-}] [OH^-]/ [PO_4^{3-}] = K_w/K_{a3} = 2.38 \times 10^{-2}$
- Thus $x^2/C_s = 2.38 \times 10^{-2}$ $\quad x = [OH^-] = 4.9 \times 10^{-2}$
- $[H_3O^+] = K_w/[OH^-] = 2.0 \times 10^{-13}$M
- **pH = 12.69 (2 sig fig)**
- Note: Check assumptions

Titrations

- **Example: Determination of HCl concentration by titration with NaOH**


20.0 mL of
acid
solution
Pipet
Initial volume
reading
Buret
Standard
NaOH
solution
20.0 mL of
acid
solution
Final volume
reading
Neutralized
solution
(indicator
has changed
color)

The Theory of Indicator Behavior

- pH-sensitive dyes have long been used as indicators.
- Normally, the basic form (In) on the dye has a colour different from the acid form, HIn:
- $HIn + H_2O \rightleftharpoons H_3O^+ + In^-$
- $In + H_2O \rightleftharpoons OH^- + HIn^+$

$$K_a = \frac{[H_3O^+][In^-]}{[HIn]} \longrightarrow \quad (1)$$

$$\frac{[H_3O^+][OH^-]}{K_a} = \frac{[OH][HIn^+]}{[In]}$$

$$[H_3O^+] = \frac{Ka[HIn^+]}{[In]} \longrightarrow \quad (2)$$

We can see that both equilibrium constant expressions above can be written:

$$[H_3O^+] = \frac{K_a[HIn]}{[In^-]} \longrightarrow \quad (3)$$

$$K_w = K_a K_b = [H_3O^+][OH^-]$$

$$\frac{[H_3O^+][OH^-]}{K_a} = \frac{[OH^-][HIn^+]}{[In]}$$

$$[H_3O^+] = \frac{Ka\,[HIn^+]}{[In]} \longrightarrow \quad (4)$$

Therefore, the $[H_3O^+]$ determines the ratio of the acid/conjugate base form of the indicator.

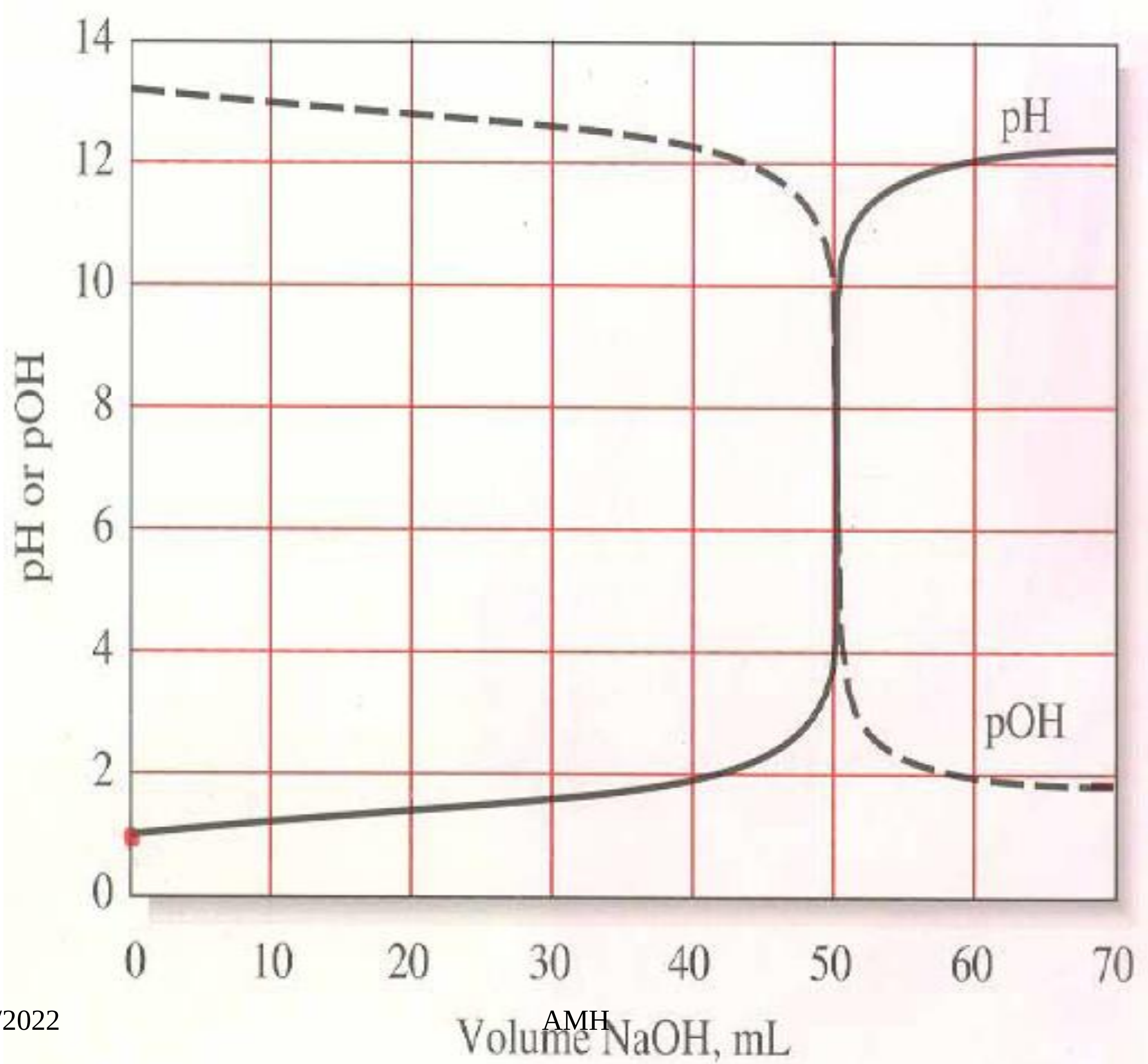
To see the colour of a particular form (acid or base) of the indicator, that form must be present at tenfold higher concentration.

- **For example,**
- **To see the colour of acid form, the ratio [HIn]/[In] must be equal to or greater than 10.**
- **To see the colour of base form, the ratio [HIn]/[In] must be equal to or less than 0.1.**
- **This means that:**
- **For acid color $[H_3O^+] > K_a (10/1)$**
- **For base color $[H_3O^+] < K_a (1/10)$**

- **Hence: indicator range = pKa ± 1, and the pH change in the area of the equivalence point must match this range or at least overlap it significantly to use this indicator for endpoint detection.**

Titration Curves

- **The X-axis units are always reagent or titrant volume.**
- **The Y-axis may be in increments of analyte reacted or a p-function such as pH (s-curve).**
- **The equivalence point is characterized by large changes in the relative concentrations of the reagent and analyte.**



14
12
10
8
6
4
2
0
pH or pOH
pH
pOH
0
10
20
30
40
50
60
70
Volume NaOH, mL

The Titration of a Strong Acid with a Strong Base

- **Example: Determination of HCl concentration by titration with NaOH**
- $NaOH + HCl \longrightarrow NaCl + H_2O$
- **moles =** $\mathbf{C_{NaOH}V_{NaOH} = C_{HCl}V_{HCl}}$

1. H_3O+ in the titration medium has two sources

- a. From the H_2O solvent
- b. From the acid solute - usually this is in great excess relative contribution from water because the Kw is so small
- c. The mass balance equation describing this situation is:
- $[H_3O^+] = C_{HCl} + [OH^-] = C_{HCl}$
- d . The same is true for a strong base and we can write :
- $[OH^-] = C_{NaOH} + [H_3O^+] = C_{NaOH}$

2. Before the equivalence point

- **We calculate the pH of the titration medium from the concentration of unreacted strong acid:**

$$\left[H_3O^+\right] = \frac{\text{Moles of}\left[H_3O^+\right] - \text{Moles of }[OH^-]}{\text{Vacid} + \text{V base}}$$

$$pH = -\log\left[H_3O^+\right]$$

3. At the equivalence point

- **All of the acid has reacted with the titrant base.**
- **For a strong acid titrated with a strong base, the salt is a strong electrolyte and therefore completely dissociated.**
- **It does not react with H_2O. The resulting solution is neutral (pH = 7.00) because:**
- $HCl + NaOH \rightleftharpoons H_2O + Na^+ + Cl^-$
- $2\ H_2O \rightleftharpoons H_3O^+ + OH^-$

4. **After the equivalence point**

- **we calculate the pH of the titration medium from the concentration of unreacted strong base:**

$$\left[OH^-\right] = \frac{K_w}{\left[H_3O^+\right]} = \frac{\text{moles of }\left[OH^-\right]\text{added} - \text{moles of }\left[H_3O^+\right]}{\text{Vacid} + \text{Vbase}}$$

Derive a curve for the titration of 50.00 mL of 0.0500 M HCl with 0.1000 M NaOH.

Initial Point

At the outset, the solution is 0.0500 M in H_3O^+, and

$$pH = -\log [H_3O^+] = -\log 0.0500 = 1.30$$

After Addition of 10.00 mL of Reagent

The hydronium ion concentration decreases as a result of both reaction with the base and dilution. Thus, the analytical concentration of HCl is

$$c_{HCl} = \frac{\text{no. mmol HCl remaining after addition of NaOH}}{\text{total volume solution}}$$

$$= \frac{\text{original no. mmol HCl} - \text{no. mmol NaOH added}}{\text{total volume solution}}$$

$$= \frac{(50.00 \text{ mL} \times 0.0500 \text{ M}) - (10.00 \text{ mL} \times 0.1000 \text{ M})}{50.00 \text{ mL} + 10.00 \text{ mL}}$$

$$= \frac{(2.500 \text{ mmol} - 1.000 \text{ mmol})}{60.00 \text{ mL}} = 2.500 \times 10^{-2} \text{ M}$$

$$[H_3O^+] = 2.50 \times 10^{-2}$$

$$pH = -\log [H_3O^+] = -\log (2.500 \times 10^{-2}) = 1.602$$

Additional points defining the curve in the region before the equivalence point are obtained in the same way. The results of such calculations are shown in the second column of Table 10-3.

Equivalence Point

At the equivalence point, neither HCl nor NaOH is in excess, and so the concentrations of hydronium and hydroxide ions must be equal. Substituting this equality into the ion-product constant for water yields

$$[H_3O^+] = \sqrt{K_w} = \sqrt{1.00 \times 10^{-14}} = 1.00 \times 10^{-7}$$

$$pH = -\log (1.00 \times 10^{-7}) = 7.00$$

After Addition of 25.10 mL of Reagent

The solution now contains an excess of NaOH, and we can write

$$c_{NaOH} = \frac{25.10 \times 0.1000 - 50.00 \times 0.0500}{75.10} = 1.33 \times 10^{-4}\ M$$

and the equilibrium concentration of hydroxide ion is

$$[OH^-] = c_{NaOH} = 1.33 \times 10^{-4}\ M$$

$$pOH = -\log (1.33 \times 10^{-4}) = 3.88$$

and

$$pH = 14.00 - 3.88 = 10.12$$

Volume of NaOH, mL	pH: 50.00 mL of 0.0500 M HCl with 0.1000 M NaOH	pH: 50.00 mL of 0.000500 M HCl with 0.001000 M NaOH
0.00	1.30	3.30
10.00	1.60	3.60
20.00	2.15	4.15
24.00	2.87	4.87
24.90	3.87	5.87
25.00	7.00	7.00
25.10	10.12	8.12
26.00	11.12	9.12
30.00	11.80	9.80

Calculate the pH during the titration of 50.00 mL of 0.0500 M NaOH with 0.1000 M HCl after the addition of the following volumes of reagent: (a) 24.50 mL, (b) 25.00 mL, (c) 25.50 mL.

(a) At 24.50 mL added, $[H_3O^+]$ is very small and cannot be computed from stoichiometric considerations but $[OH^-]$ can be obtained readily

$$[OH^-] = c_{NaOH} = \frac{\text{original no. mmol NaOH} - \text{no. mmol HCl added}}{\text{total volume of solution}}$$

$$= \frac{(50.00 \times 0.0500 - 24.50 \times 0.1000)}{50.00 + 24.50} = 6.71 \times 10^{-4}$$

$$[H_3O^+] = K_w/(6.71 \times 10^{-4}) = (1.00 \times 10^{-14})/(6.71 \times 10^{-4})$$
$$= 1.49 \times 10^{-11}$$
$$pH = -\log(1.49 \times 10^{-11}) = 10.83$$

(b) This is the equivalence point where $[H_3O^+] = [OH^-]$

$$[H_3O^+] = \sqrt{K_w} = \sqrt{1.00 \times 10^{-14}} = 1.00 \times 10^{-7}$$
$$pH = -\log(1.00 \times 10^{-7}) = 7.00$$

(c) $$[H_3O^+] = c_{HCl} = \frac{(25.50 \times 0.1000 - 50.00 \times 0.0500)}{75.50}$$
$$= 6.62 \times 10^{-4}$$
$$pH = -\log(6.62 \times 10^{-4}) = 3.18$$